SEMESTER - II

COURSE: MAJOR – 2(MJ-2) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: MATHEMATICAL PHYSICS-I **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

COURSE OBJECTIVES:

This course aims to enable the students to acquire the key concepts of the general properties of matter, the motion of a particle under central force field, oscillations and special theory of relativity.

COURSE OUTCOMES:

1. Learn about the behavior of physical bodies around us in daily life.
2. Understand the dynamics of planetary motion.
3. Build a foundation of various applied field in science and technology.
4. Develop the analytical thinking on Mechanics in order to understand the response of the classical systems to external forces.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Vector Calculus (15 Lectures): Scalar and Vector fields, Vector Differentiation: Directional derivatives and normal derivative. Gradient of a scalar field and its geometrical interpretation. Divergence and curl of a vector field and their physical interpretation. Line, surface and volume integrals. Gauss' divergence theorem, Green's and Stokes Theorems and their applications. Derivation of Gradient, Divergence, Curl and Laplacian in Cartesian, Spherical and Cylindrical Coordinate Systems.

UNIT-II:

Differential Equation (15 Lectures): First order differential Equations, homogeneous, non-homogeneous, exact and inexact differential equations and Integrating Factors. Homogeneous Equations with constant coefficients. Cauchy-Euler differential equation and simultaneous differential equations of First and Second order. Solutions to partial differential equations (2 or 3 independent variables) using separation of variables: Laplace's Equation in problems of rectangular geometry. Solution of wave equation for vibrational modes of a stretched string, rectangular and circular membranes. Solution of 1D heat flow equation. (Wave/Heat equation not to be derived).

UNIT-III:

Complex Analysis (15 Lectures): Brief Revision of Complex Numbers and their Graphical Representation. Euler's formula, De-Moivre's theorem, Roots of Complex Numbers. Functions of Complex Variables. Analyticity and Cauchy-Riemann Equations. Examples of analytic functions. Singularities: poles, removable singularity, essential singularity, branch points, branch cut. Integration of a function of a complex variable. Cauchy-Goursat Theorem, Cauchy's Inequality. Cauchy's Integral formula. Simply and multiply connected region. Laurent and Taylor's expansion. Residues and Residue Theorem. Application of Contour Integration in solving Definite Integrals.

SUGGESTED READINGS:

1. Mathematical Methods for Physicists, G.B. Arfken, H.J. Weber, F.E. Harris, 2013, 7th Edn., Academic Press/Elsevier.
2. Mathematical Physics, H. K. Dass and Rama Verma, S. Chand & Company Pvt. Ltd.
3. Modern Mathematical Methods for Physicists and Engineers, C.D. Cantrell, 2011, Cambridge University Press.
4. Mathematical Methods for Physicis & Engineers, K.F. Riley, M.P. Hobson, S.J. Bence, 3rd Ed., 2006, Cambridge University Press.
5. Mathematical physics by B.S. Rajput, A Pragati Edition.
6. Advanced Methods of Mathematical Physics by R.S. Kaushal and D. Parashar, Narosa Publishing House, New Delhi.
7. Mathematical Physics by B. D. Gupta, Vikash Publishing House Pvt. Ltd.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Measurements of length (or diameter) using Vernier Caliper.
2. Measurement of Thickness using Screw Gauge and Travelling Microscope.

3. Measurement of thickness of thin transparent plates using Spherometer.
4. Determination of radius of curvature using Spherometer.
5. Measurement of volume and density of (A) a solid cylinder and (B) hollow cylinder (pipe), using Vernier calipers and Screw Gauge.
6. Use of a multimeter for measuring (a) Resistances, (b) AC and DC Voltages, (c) DC Current, (d) Capacitances, and (e) Checking electrical fuses.

SUGGESTED READINGS:

1. Advanced Practical Physics for students, B. L. Flint and H.T. Worsnop, 1971, Asia Publishing House.
2. A Text Book of Practical Physics, I. Prakash & Ramakrishna, 11th Edn, 2011, Kitab Mahal. Publications.

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